# Mohd Abdul Nawaz

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## PROFESSIONAL SUMMARY

B.Tech Computer Science student at IIIT Guwahati with a strong focus on Data Science and Machine Learning. Experienced in building end-to-end ML workflows, statistical data cleaning, predictive modeling and model interpretability. Hackathon award-winner (Start-Up Med Hackathon — 2nd Prize; UIDAI Hackathon — 6th Prize) with hands-on work on healthcare risk prediction and large-scale societal data analytics.

## EDUCATION

**B.Tech in Computer Science & Engineering | IIIT Guwahati**
2024 – 2028 | CPI: 7.30

**Class XII (ICSE) | S.T. George's Grammar Boys School, Telangana**
2023 | 94.25%

**Class X (ICSE) | Narayana Junior College, Telangana**
2021 | 88.9%

## TECHNICAL SKILLS

**Programming Languages:**
- Python, C++, Java, C, SQL

**Data Science & Analysis:**
- Pandas, NumPy, Matplotlib, Seaborn, Plotly

**Machine Learning:**
- scikit-learn — Linear & Robust (Huber) Regression, Decision Trees, Random Forest, AdaBoost, Logistic Regression
- Hyperparameter tuning — GridSearchCV, RandomizedSearchCV, Cross-Validation

**Deep Learning & NLP:**
- TensorFlow / Keras — CNNs, MLPs, transfer learning (computer vision)
- NLP — NLTK, spaCy, Word2Vec, GloVe, Sentence Transformers, Hugging Face Transformers

**Data Processing & Feature Engineering:**
- Missing-value handling (MICE / IterativeImputer), outlier treatment (RobustScaler), skew-aware transformations, time-series forecasting

**Model Interpretability:**
- SHAP, decision-path analysis

**Tools & Platforms:**
- Jupyter Notebook, Google Colab, Git, GitHub

## PROJECTS

**Start-Up Med Hackathon — ICU Patient Risk Prediction (2nd Prize)**
- Developed a clinical decision-support model predicting ICU patient risk on merged patient and periodic-vitals data keyed by patient stay ID.
- Engineered clinically-safe imputation — linear interpolation, expanding running averages and special handling for expired patients — to clean noisy and missing vital-sign streams.
- Built a pruned Decision Tree classifier with tuned F2-score, achieving 92% precision stable across diverse patient stays.
- Added interpretability through SHAP analysis and per-patient decision-path explanations for bedside use.

**UIDAI Hackathon — Aadhaar Enrolment & Updates Trend Analysis (6th Prize)**
- Analyzed large-scale Aadhaar enrolment, demographic-update and biometric datasets to uncover seasonal, state, district and age-group trends.
- Performed univariate, bivariate and trivariate EDA and built time-series forecasting models (SARIMA, WindowAverage) evaluated with MAE, RMSE and MAPE (best RMSE ≈ 232).
- Delivered actionable strategies (e.g., month-wise staffing aligned to seasonal peaks) supporting informed decision-making.

**Important Functions — Reusable Machine-Learning Workflows Library**
- Designed a reusable Python library automating end-to-end ML pipelines (cleaning → training → evaluation).
- Implemented MICE-based missing-value imputation with skew-aware log transformations that are reversed to the original scale.
- Built modular classical-ML workflows (Decision Tree, Random Forest, AdaBoost, Gradient Boosting, XGBoost, Logistic Regression) with GridSearch/RandomizedSearch, CCP pruning and outlier-robust Huber regression.
- Added deep learning & NLP — CNNs with transfer learning and MLP grid search (TensorFlow/Keras) for computer vision, plus NLP sentiment analysis using Word2Vec/GloVe embeddings, sentence transformers and Hugging Face (Flan-T5) models.

## CERTIFICATIONS & ACHIEVEMENTS

- Data Science using Generative AI — Certificate of Completion
- 2nd Prize — Start-Up Med Hackathon
- 6th Prize — UIDAI Hackathon